

Name _____

Total extra pages _____

Use Password **IVEGOTTHIS** to start the exam.

-
- You may use as many pages as you need to show your work.
 - Please write your name on **every page** that you use to show your work.
 - You may use the front and back of the paper.
 - Clearly **label** your work with the problem number you are working on.
 - **Record which problems you worked** on each page (front and back).
 - When you have completed the exam, count the number of extra pages you used, then **label each page** in order of the problem number.
 - You will **staple** your pages together with **this page on top** when you complete your exam.

Problem number	Point Adjustment	Out of	Notes
1		1	
2		1.4	
3		1	
4		1.2	
5		1.4	
6		1.5	
7		1	
8		1.5	
Total adjustment		10	ALEKS SCORE: Adjusted total score:

Name_____

Page ____ of ____

This page shows work for problems_____

Do not write your work or solutions above this line.

Basic Differentiation Rules	Basic Integration Rules
$\frac{d}{dx} C = 0$	$\int 0 \, dx = C$
$\frac{d}{dx} (kx) = k$	$\int k \, dx = kx + C$
$\frac{d}{dx} [kf(x)] = kf'(x)$	$\int [kf(x)] \, dx = k \int f(x) \, dx$
$\frac{d}{dx} [f(x) \pm g(x)] = f'(x) \pm g'(x)$	$\int [f(x) \pm g(x)] \, dx = \int f(x) \, dx \pm \int g(x) \, dx$
$\frac{d}{dx} x^n = nx^{n-1}$	$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \, n \neq -1 \quad (\text{Power Rule})$
$\frac{d}{dx} \sin x = \cos x$	$\int \cos x \, dx = \sin x + C$
$\frac{d}{dx} \cos x = -\sin x$	$\int \sin x \, dx = -\cos x + C$
$\frac{d}{dx} \tan x = \sec^2 x$	$\int \sec^2 x \, dx = \tan x + C$
$\frac{d}{dx} \cot x = -\csc^2 x$	$\int \csc^2 x \, dx = -\cot x + C$
$\frac{d}{dx} \sec x = \sec x \tan x$	$\int \sec x \tan x \, dx = \sec x + C$
$\frac{d}{dx} \csc x = -\csc x \cot x$	$\int \csc x \cot x \, dx = -\csc x + C$
$\frac{d}{dx} e^x = e^x$	$\int e^x \, dx = e^x + C$
$\frac{d}{dx} \ln x = \frac{1}{x}, \, x > 0$	$\int \frac{1}{x} \, dx = \ln x + C$

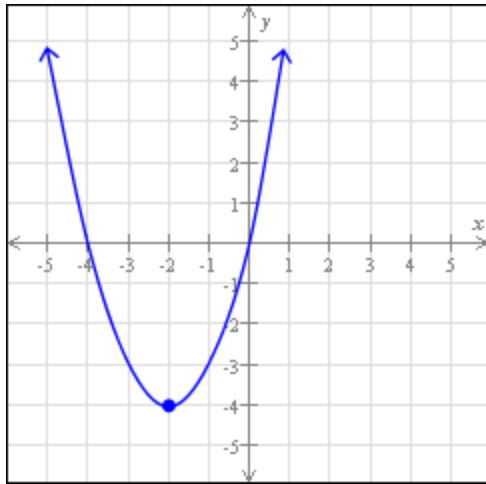
Other tips:

- A function is concave up on the interval (a, b) if $f''(x) > 0$ for all x in (a, b) .
- A function is concave down on the interval (a, b) if $f''(x) < 0$ for all x in (a, b) .
- If $f(x)$ is a continuous function on the interval $[a, b]$ and $F'(x) = f(x)$, then $\int_a^b f(x) \, dx = F(b) - F(a)$.
- If $\lim_{x \rightarrow a^\pm} f(x) = \pm\infty$, then $x = a$ is a vertical asymptote.
- If $\lim_{x \rightarrow \pm\infty} f(x) = c$, where c is a finite number, then $y = c$ is a horizontal asymptote.
- The point-slope equation of a line with slope m and through the point (x_1, y_1) is $y = y_1 + m(x - x_1)$.
- $\frac{1}{x^n} = x^{n-1}$ and $x^{1/m} = \sqrt[m]{x}$.
- You might need the product rule: $\frac{d}{dx} f(x)g(x) = f'(x)g(x) + g'(x)f(x)$

Student Name: _____

Class Name: **Calculus 1 Beta**Number of Questions: **8**Instructor Name: **Question 1 of 8**

Here is a graph of the function g .



Use the graph to find the following.

If there is more than one answer, separate them with commas.

All values at which g has a local minimum:

All local minimum values of g :

Question 2 of 8

Let the function f be defined as follows.

$$f(x) = x^4 - 8x^3 + 18x^2 - 5$$

Find the critical numbers (critical points) of the function. Use exact values (not decimal approximations). If there is more than one number, separate them with commas. Click on "None" if applicable.

Critical numbers of f : _____

Question 3 of 8

Consider the function defined by the equation below.

$$y = 2x^2 - x - 3$$

Find the absolute minimum and absolute maximum values of y on the interval $[-2, 3]$. Give exact answers, not decimal approximations. If there is no absolute minimum or absolute maximum value, write "None".

Absolute minimum value: _____

Absolute maximum value: _____

Question 4 of 8

Find the following limit. If necessary, select the most informative answer from $-\infty$, ∞ , and "Does Not Exist."

$$\lim_{x \rightarrow \infty} \frac{x - 3}{5 - 8x} = \underline{\hspace{2cm}}$$

Question 5 of 8

Consider the function f given below.

$$f(x) = (3x^3 - 1)e^{3x}$$

Find $L(x)$, the linearization of f at the value $a=0$.

$$L(x) =$$

Question 6 of 8

Let the function f be defined as follows.

$$f(x) = \frac{1}{2}x^4 - 5x^3 + 12x^2 + 5x + 1$$

- (a) Determine the interval(s) in the domain of f on which the function is concave up, and the interval(s) in the domain of f on which the function is concave down. Use exact values in your answers (not decimal approximations).

Write your answers as an interval or list of intervals. When writing a list of intervals, make sure to separate them with commas and to use as few intervals as possible. Write "None" if applicable.

Interval(s) on which f is concave up: _____

Interval(s) on which f is concave down: _____

- (b) Find all the inflection points (points of inflection) of the function f . Write the points as ordered pairs. Use exact values in your answers (not decimal approximations).

If there is more than one point, separate them with commas. Write "None" if applicable.

All inflection points of f : _____

Question 7 of 8

Find the integral.

$$\int (6z^6 + 5z^2 + 4) dz = \underline{\hspace{2cm}}$$

Question 8 of 8

Evaluate the integral. Give an exact answer and not a decimal approximation.

$$\int_{-1}^2 (5z^3 + 3z - 8) dz = \underline{\hspace{2cm}}$$

Class Name: **Calculus 1 Beta**Number of Questions: **8****Question 1 of 8**All values at which g has a local minimum: -2 All local minimum values of g : -4 **Question 2 of 8**Critical numbers of f : $0, 3$ **Question 3 of 8**Absolute minimum value: $y = -\frac{25}{8}$ Absolute maximum value: $y = 12$ **Question 4 of 8**

$$\lim_{x \rightarrow \infty} \frac{x - 3}{5 - 8x} = -\frac{1}{8}$$

Question 5 of 8

$$L(x) = -3x - 1$$

Question 6 of 8

(a) Interval(s) on which f is concave up: $(-\infty, 1), (4, \infty)$

Interval(s) on which f is concave down: $(1, 4)$

(b) All inflection points of f : $\left(1, \frac{27}{2}\right), (4, 21)$

Question 7 of 8

$$\int (6z^6 + 5z^2 + 4) dz = \frac{6}{7}z^7 + \frac{5}{3}z^3 + 4z + C$$

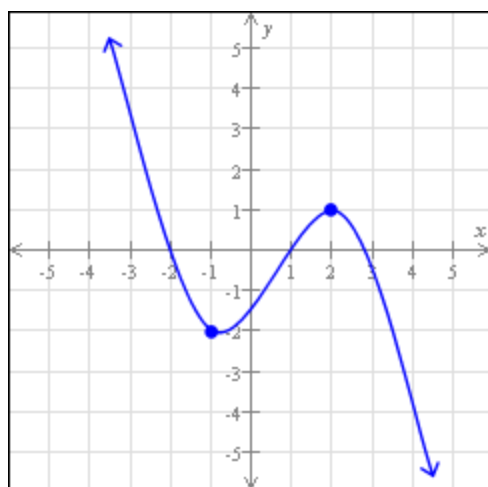
Question 8 of 8

$$\int_{-1}^2 (5z^3 + 3z - 8) dz = -\frac{3}{4}$$

Student Name: _____

Class Name: **Calculus 1 Beta**Number of Questions: **8**Instructor Name: **Ruiz, Amanda****Question 1 of 8**

Here is a graph of the function g .



Use the graph to find the following.

If there is more than one answer, separate them with commas.

All local maximum values of g :

All values at which g has a local maximum:

Question 2 of 8

Let the function f be defined as follows.

$$f(x) = x^3 - 6x^2 + 9x - 7$$

Find the critical numbers (critical points) of the function. Use exact values (not decimal approximations). If there is more than one number, separate them with commas. Click on "None" if applicable.

Critical numbers of f : _____

Question 3 of 8

Consider the function defined by the equation below.

$$y = -1 - 3x^2 + 4x$$

Find the absolute minimum and absolute maximum values of y on the interval $[-2, 2]$. Give exact answers, not decimal approximations. If there is no absolute minimum or absolute maximum value, write "None".

Absolute minimum value: _____

Absolute maximum value: _____

Question 4 of 8

Find the following limit. If necessary, select the most informative answer from $-\infty$, ∞ , and "Does Not Exist."

$$\lim_{x \rightarrow -\infty} \frac{x^9 - 2}{x^8 - 10} = \underline{\hspace{2cm}}$$

Question 5 of 8

Consider the function f given below.

$$f(x) = 4\sqrt{x} \left(e^{\frac{1}{4}x-1} \right)$$

Find $L(x)$, the linearization of f at the value $a=4$.

$$L(x) =$$

Question 6 of 8

Let the function f be defined as follows.

$$f(x) = -x^3 + 12x^2 + 7x + 2$$

- (a) Determine the interval(s) in the domain of f on which the function is concave up, and the interval(s) in the domain of f on which the function is concave down. Use exact values in your answers (not decimal approximations).

Write your answers as an interval or list of intervals. When writing a list of intervals, make sure to separate them with commas and to use as few intervals as possible. Write "None" if applicable.

Interval(s) on which f is concave up: _____

Interval(s) on which f is concave down: _____

- (b) Find all the inflection points (points of inflection) of the function f . Write the points as ordered pairs. Use exact values in your answers (not decimal approximations).

If there is more than one point, separate them with commas. Write "None" if applicable.

All inflection points of f : _____

Question 7 of 8

Find the integral.

$$\int \left(-5t^{\frac{2}{3}} + 2t^{-\frac{5}{3}} + 5t^{\frac{3}{2}} \right) dt = \underline{\hspace{2cm}}$$

Question 8 of 8

Evaluate the integral. Give an exact answer and not a decimal approximation.

$$\int_{-2}^1 \left(2x^3 - 2x^2 - 5 \right) dx = \underline{\hspace{2cm}}$$

Class Name: **Calculus 1 Beta**Number of Questions: **8****Question 1 of 8**All local maximum values of g : 1All values at which g has a local maximum: 2**Question 2 of 8**Critical numbers of f : 1, 3**Question 3 of 8**Absolute minimum value: $y = -21$ Absolute maximum value: $y = \frac{1}{3}$ **Question 4 of 8**

$$\lim_{x \rightarrow -\infty} \frac{x^9 - 2}{x^8 - 10} = -\infty$$

Question 5 of 8

$$L(x) = 3x - 4$$

Question 6 of 8

(a)

Interval(s) on which f is concave up: $(-\infty, 4)$

Interval(s) on which f is concave down: $(4, \infty)$

(b)

All inflection points of f : $(4, 158)$

Question 7 of 8

$$\int \left(-5t^{\frac{2}{3}} + 2t^{-\frac{5}{3}} + 5t^{\frac{3}{2}} \right) dt = -3t^{\frac{5}{3}} - 3t^{-\frac{2}{3}} + 2t^{\frac{5}{2}} + C$$

Question 8 of 8

$$\int_{-2}^1 (2x^3 - 2x^2 - 5) dx = -\frac{57}{2}$$

Student Name: _____

Class Name: **Calculus 1 Beta**

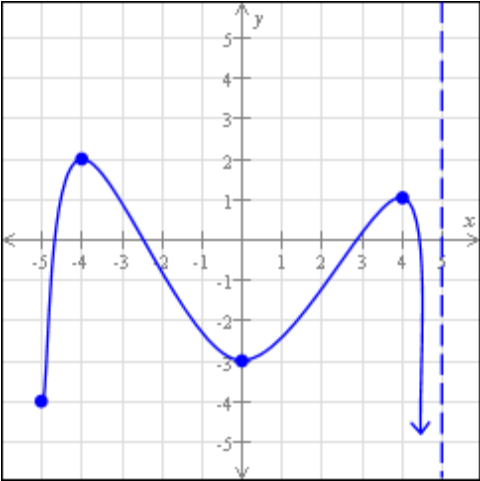
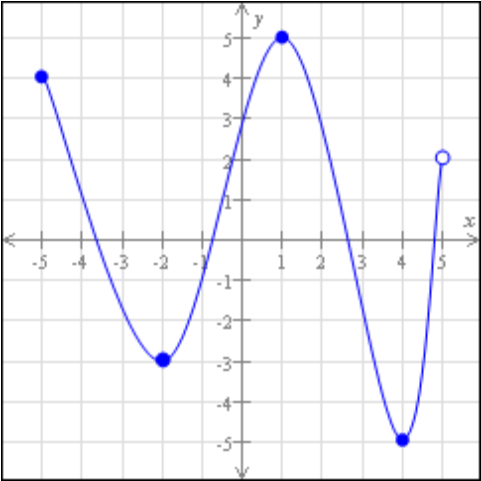
Number of Questions: **8**

Instructor Name: **Ruiz, Amanda**

Question 1 of 8

The graphs of the functions g and h are shown below. For each graph, find the absolute maximum and absolute minimum.

Assume that the dashed line shown is a vertical asymptote that the graph does not cross.

(a) Graph of g	(b) Graph of h
	
Absolute maximum of g : _____	Absolute maximum of h : _____
Absolute minimum of g : _____	Absolute minimum of h : _____

Question 2 of 8

Let the function f be defined as follows.

$$f(x) = \frac{3}{4}x^4 + 4x^3 + 6x^2 + 2$$

Find the critical numbers (critical points) of the function. Use exact values (not decimal approximations). If there is more than one number, separate them with commas. Click on "None" if applicable.

Critical numbers of f : _____

Question 3 of 8

Consider the function defined by the equation below.

$$y = 2x^2 - x - 3$$

Find the absolute minimum and absolute maximum values of y on the interval $[-2, 3]$. Give exact answers, not decimal approximations. If there is no absolute minimum or absolute maximum value, write "None".

Absolute minimum value: _____

Absolute maximum value: _____

Question 4 of 8

Find the following limit. If necessary, select the most informative answer from $-\infty$, ∞ , and "Does Not Exist."

$$\lim_{x \rightarrow \infty} \frac{x^9 - 2}{x^8 - 6} = \underline{\hspace{2cm}}$$

Question 5 of 8

Consider the function f given below.

$$f(x) = 2\sqrt{x}(e^{3x-3})$$

Find $L(x)$, the linearization of f at the value $a = 1$.

$$L(x) = \underline{\hspace{2cm}}$$

Question 6 of 8

Let the function f be defined as follows.

$$f(x) = x^4 + 8x^3 + 7x + 2$$

- (a) Determine the interval(s) in the domain of f on which the function is concave up, and the interval(s) in the domain of f on which the function is concave down. Use exact values in your answers (not decimal approximations).

Write your answers as an interval or list of intervals. When writing a list of intervals, make sure to separate them with commas and to use as few intervals as possible. Write "None" if applicable.

Interval(s) on which f is concave up: _____

Interval(s) on which f is concave down: _____

- (b) Find all the inflection points (points of inflection) of the function f . Write the points as ordered pairs. Use exact values in your answers (not decimal approximations).

If there is more than one point, separate them with commas. Write "None" if applicable.

All inflection points of f : _____

Question 7 of 8

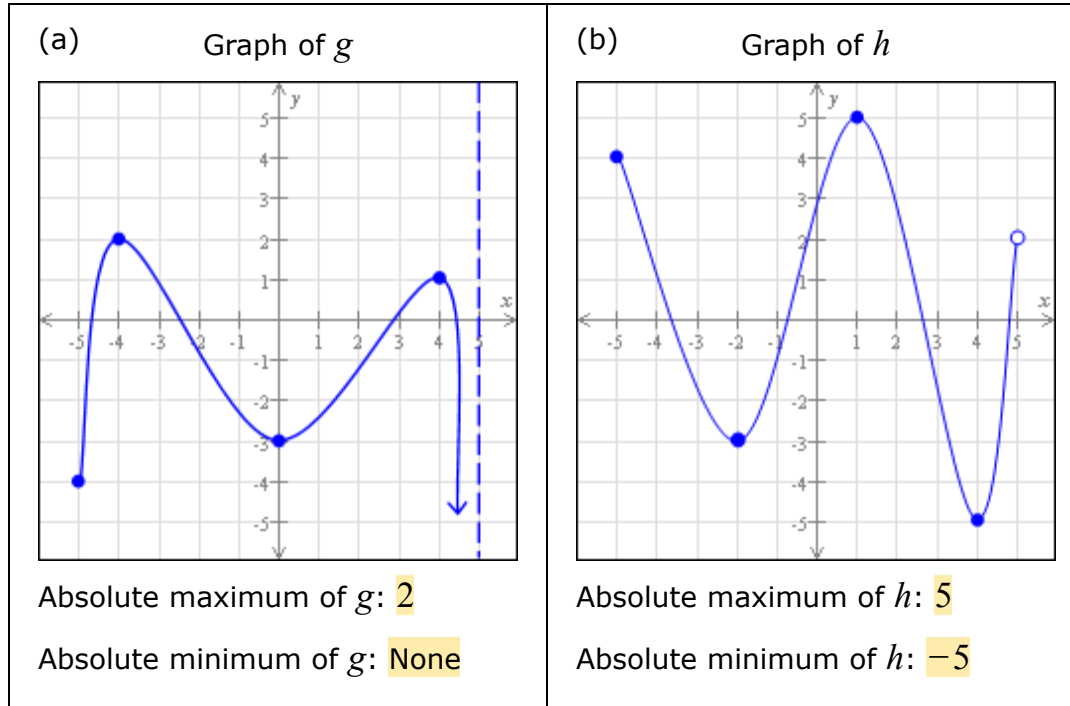
Find the integral.

$$\int \left(\frac{2}{4x^5} - \frac{3}{6x^5} + \frac{5}{5x^2} \right) dx = \underline{\hspace{2cm}}$$

Question 8 of 8

Evaluate the integral. Give an exact answer and not a decimal approximation.

$$\int_{-2}^2 (4z^2 - 3z - 3) dz = \underline{\hspace{2cm}}$$

Class Name: **Calculus 1 Beta**Number of Questions: **8****Question 1 of 8****Question 2 of 8**Critical numbers of f : $0, -2$ **Question 3 of 8**Absolute minimum value: $y = -\frac{25}{8}$ Absolute maximum value: $y = 12$

Question 4 of 8

$$\lim_{x \rightarrow \infty} \frac{x^9 - 2}{x^8 - 6} = \infty$$

Question 5 of 8

$$L(x) = 7x - 5$$

Question 6 of 8

(a) Interval(s) on which f is concave up: $(-\infty, -4), (0, \infty)$

Interval(s) on which f is concave down: $(-4, 0)$

(b) All inflection points of f : $(-4, -282), (0, 2)$

Question 7 of 8

$$\int \left(\frac{2}{4x^5} - \frac{3}{6x^5} + \frac{5}{5x^2} \right) dx = \frac{20x^{\frac{7}{5}}}{7} - 15x^{\frac{2}{5}} + \frac{10x^{\frac{7}{2}}}{7} + C$$

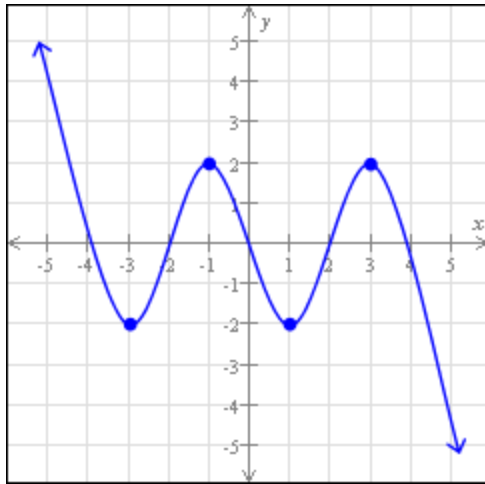
Question 8 of 8

$$\int_{-2}^2 (4z^2 - 3z - 3) dz = \frac{28}{3}$$

Student Name: _____

Class Name: **Calculus 1 Beta**Number of Questions: **8**Instructor Name: **Ruiz, Amanda****Question 1 of 8**

Here is a graph of the function h .



Use the graph to find the following.

If there is more than one answer, separate them with commas.

All values at which h has a local maximum:

All local maximum values of h :

Question 2 of 8

Let the function f be defined as follows.

$$f(x) = 3x^4 - 8x^3 + 6x^2 + 1$$

Find the critical numbers (critical points) of the function. Use exact values (not decimal approximations). If there is more than one number, separate them with commas. Click on "None" if applicable.

Critical numbers of f : _____

Question 3 of 8

Consider the function defined by the equation below.

$$y = 3x^2 + 8x - 3$$

Find the absolute minimum and absolute maximum values of y on the interval $[-3, 1]$. Give exact answers, not decimal approximations. If there is no absolute minimum or absolute maximum value, write "None".

Absolute minimum value: _____

Absolute maximum value: _____

Question 4 of 8

Find the following limit. If necessary, select the most informative answer from $-\infty$, ∞ , and "Does Not Exist."

$$\lim_{x \rightarrow -\infty} \frac{x^9 - 2}{x^8 - 10} = \underline{\hspace{2cm}}$$

Question 5 of 8

Consider the function f given below.

$$f(x) = 4\sqrt{x} \left(e^{\frac{1}{4}x-1} \right)$$

Find $L(x)$, the linearization of f at the value $a=4$.

$$L(x) =$$

Question 6 of 8

Let the function f be defined as follows.

$$f(x) = x^3 + 15x^2 + 3x + 5$$

- (a) Determine the interval(s) in the domain of f on which the function is concave up, and the interval(s) in the domain of f on which the function is concave down. Use exact values in your answers (not decimal approximations).

Write your answers as an interval or list of intervals. When writing a list of intervals, make sure to separate them with commas and to use as few intervals as possible. Write "None" if applicable.

Interval(s) on which f is concave up: _____

Interval(s) on which f is concave down: _____

- (b) Find all the inflection points (points of inflection) of the function f . Write the points as ordered pairs. Use exact values in your answers (not decimal approximations).

If there is more than one point, separate them with commas. Write "None" if applicable.

All inflection points of f : _____

Question 7 of 8

Find the integral.

$$\int \left(-4t^6 - 3t^4 + 5 \right) dt = \underline{\hspace{2cm}}$$

Question 8 of 8

Evaluate the integral. Give an exact answer and not a decimal approximation.

$$\int_{-2}^0 \left(8x^2 + 6x - 5 \right) dx = \underline{\hspace{2cm}}$$

Class Name: **Calculus 1 Beta**Number of Questions: **8****Question 1 of 8**All values at which h has a local maximum: $-1, 3$ All local maximum values of h : 2 **Question 2 of 8**Critical numbers of f : $0, 1$ **Question 3 of 8**Absolute minimum value: $y = -\frac{25}{3}$ Absolute maximum value: $y = 8$ **Question 4 of 8**

$$\lim_{x \rightarrow -\infty} \frac{x^9 - 2}{x^8 - 10} = -\infty$$

Question 5 of 8

$$L(x) = 3x - 4$$

Question 6 of 8

(a)

Interval(s) on which f is concave up: $(-5, \infty)$

Interval(s) on which f is concave down: $(-\infty, -5)$

(b)

All inflection points of f : $(-5, 240)$

Question 7 of 8

$$\int (-4t^6 - 3t^4 + 5) dt = -\frac{4}{7}t^7 - \frac{3}{5}t^5 + 5t + C$$

Question 8 of 8

$$\int_{-2}^0 (8x^2 + 6x - 5) dx = -\frac{2}{3}$$